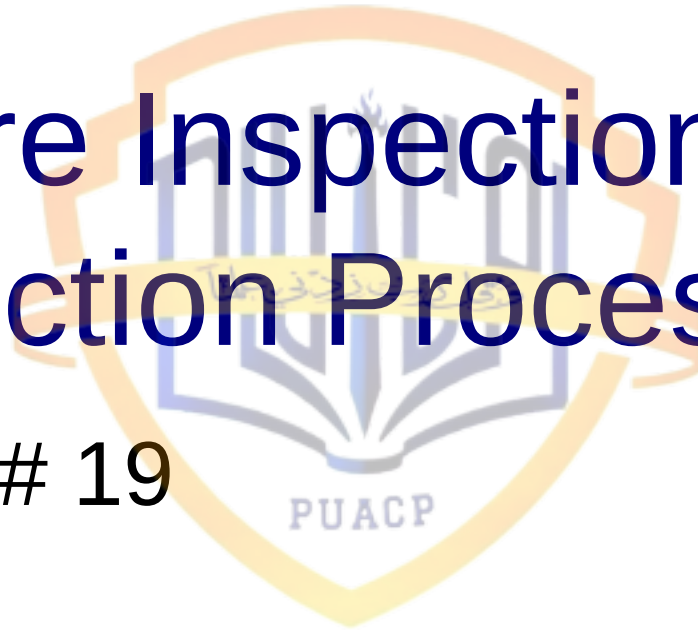




Software Inspections – 5

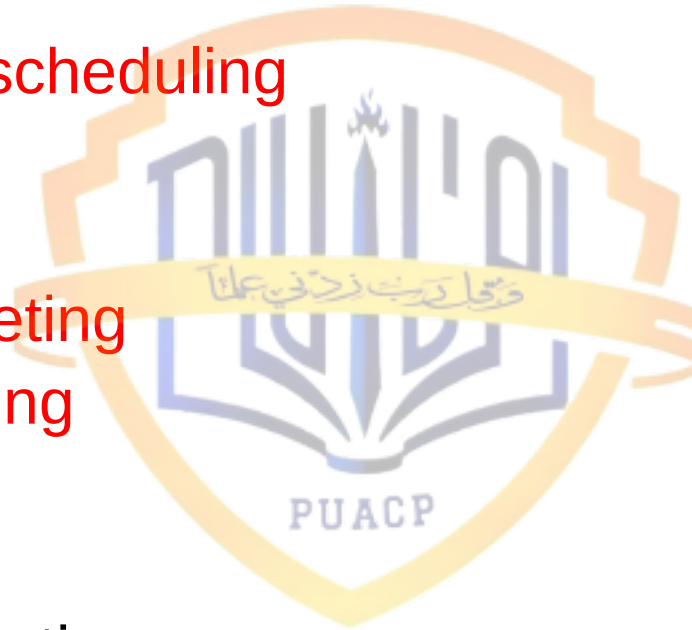
– Inspection Process

Lecture # 19

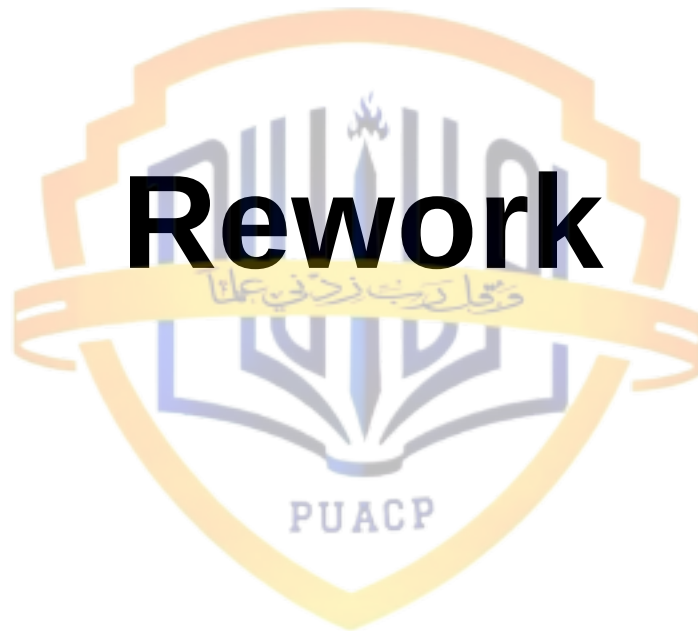


Recap and Today's Lecture: Inspection Process

- Planning and scheduling
- Overview
- Preparation
- Inspection meeting
- Analysis meeting
- Rework
- Follow-up
- Prevention meeting
- Data recording and reports
- Inspection process monitoring



Rework



Rework

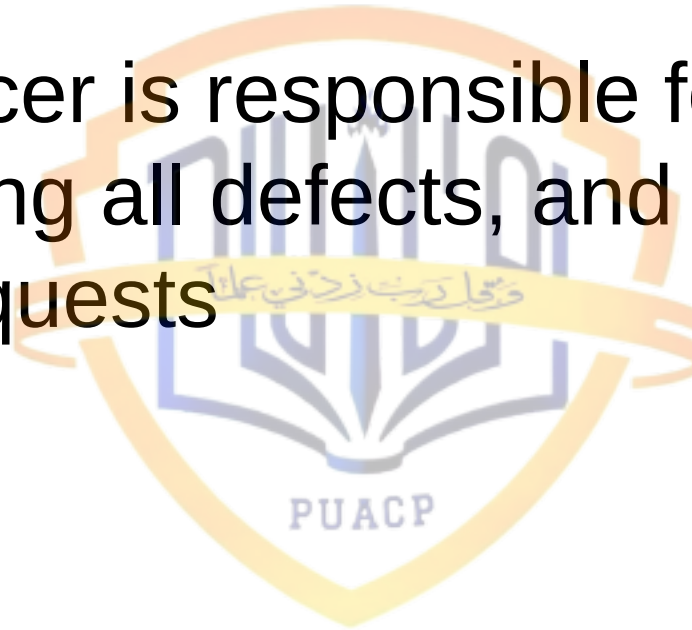
- Fixes identified defects and resolves any open issues noted during the inspection
- In some cases, the repair may require a Change request to be written because of the nature or impact of the defect

Open Issues in Rework

- The open issue is accepted as a defect and repaired
- The open issue becomes a change request for further investigation
- The open issue was incorrectly identified as potential defect and is closed as a non-defect

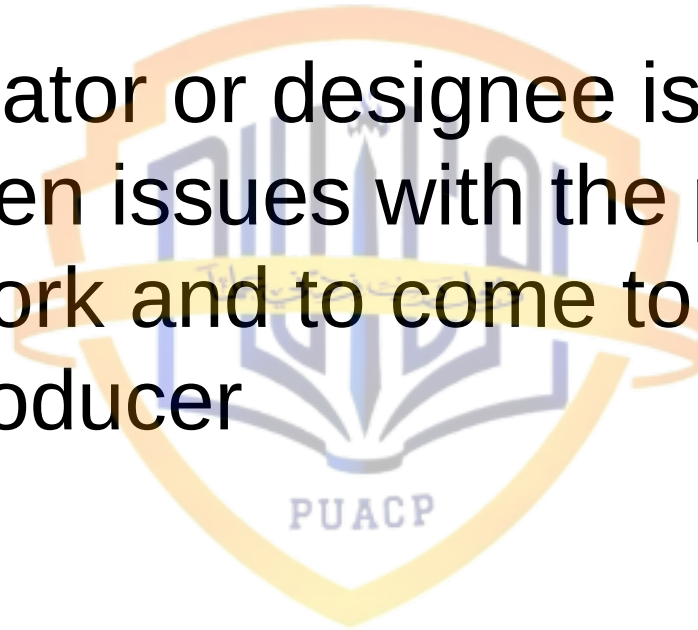
Rework: Responsibility

- The producer is responsible for all open issues, fixing all defects, and writing any change requests



Rework: Other Roles

- The moderator or designee is assigned to discuss open issues with the producer during rework and to come to concurrence with the producer



Rework: Entry Criteria

- The list of defects and open issues is provided to the producer for resolution
- The moderator or someone assigned meets with the producer to review rework and open issues
- The inspection report is completed, is on file, and available

Rework: Tasks

- The producer repairs accepted defects identified during the inspection meeting
- The producer resolves any open issues
- The moderator meets with the producer to discuss resolutions of open issues
- Change requests are written for any open issues or defects not resolved during the rework activity
- Either the minor defect list or marked work products with minor defects noted are used to repair the minor defects

Rework: Validation/Verification - 1

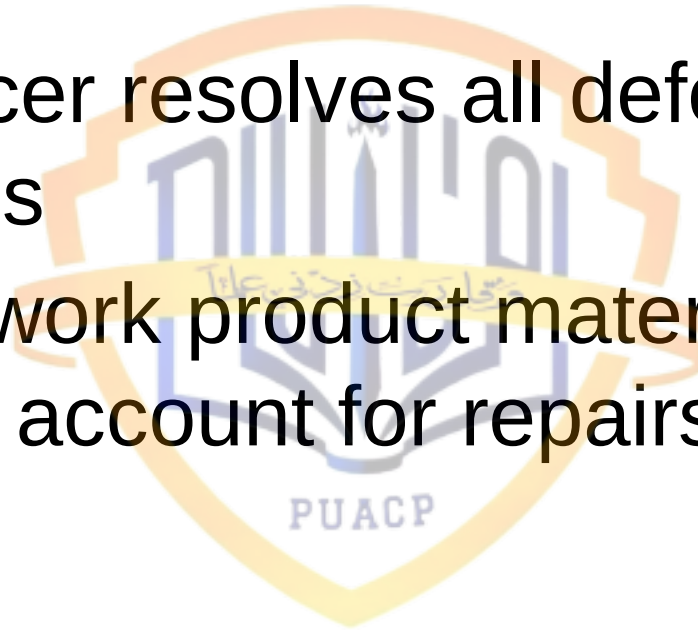
- The follow-up activity is scheduled; where the rework will be verified by the moderator or assigned designee
- SQA has reviewed sample results of this activity in the project

Rework: Validation/Verification - 2

- Data gathered during this activity
 - How much time was spent in rework
 - How many open issues were resolved and accepted as defects
 - How many open issues became submitted change requests

Rework: Exit Criteria

- The producer resolves all defects and open issues
- Inspected work product materials are updated to account for repairs

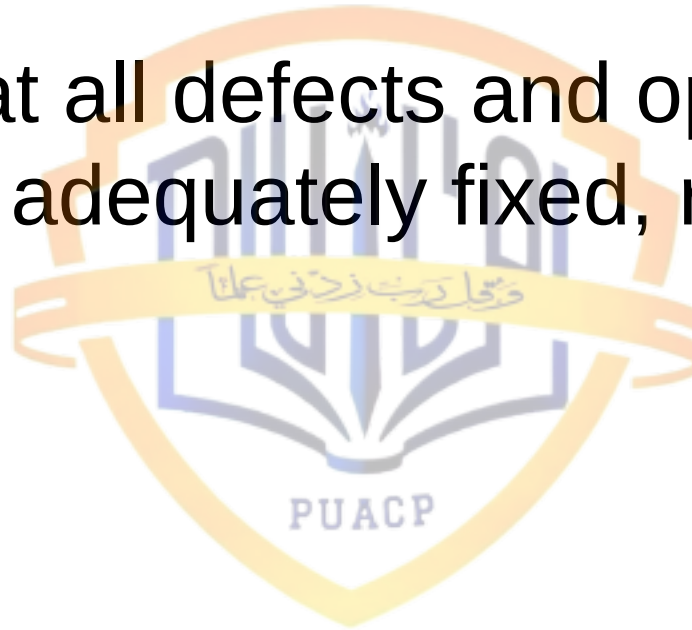




Follow-Up

Follow-Up

- Verifies that all defects and open issues have been adequately fixed, resolved, and closed out

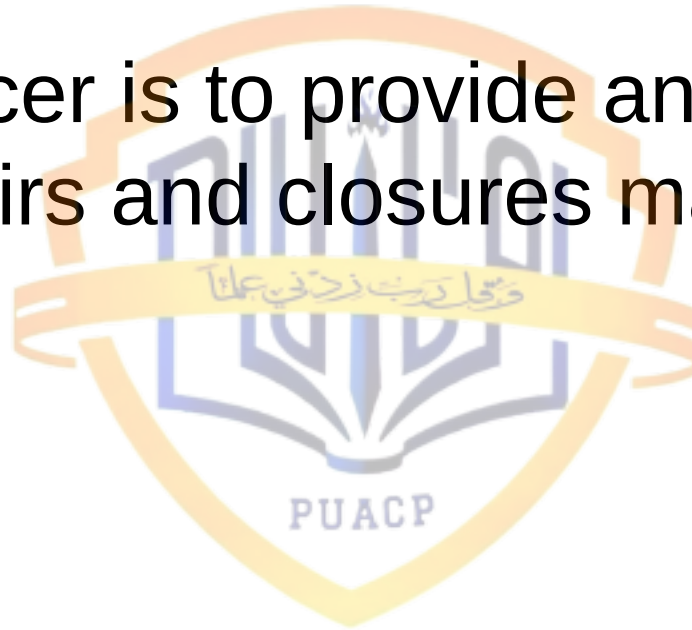


Follow-up: Responsibility

- The moderator is the individual primarily responsible for reviewing repairs. The moderator will also review the producer's decisions on repairs and change requests. The moderator may delegate some or all of this responsibility

Follow-up: Other Roles

- The producer is to provide an explanation of the repairs and closures made



Follow-up: Entry Criteria

- Rework of defects has been completed; i.e., fixed or identified with a decision to not fix
- The producer has completed the rework for defects and open issues resolved to be defects
- Change requests are written for any defects or open issues not resolved
- The moderator concurs with the producer's decisions on defects, open issues, and change requests

Follow-up: Tasks - 1

- The moderator and producer discuss and agree on compliance with respect to defects and open issues
- In case of disagreement, the issue would be resolved by the project lead
- The producer updates the work product to reflect the fixes to defects found and open issues accepted as defects

Follow-up: Tasks - 2

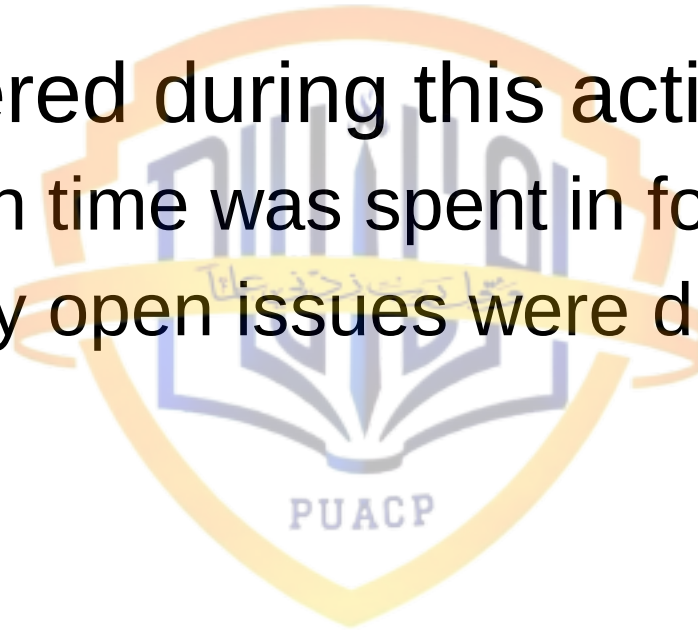
- The producer writes any change requests that may be required
- The moderator completes the inspection report and marks the inspection as closed

Follow-up: Validation/Verification - 1

- The moderator concurs with the defect repairs and open issue closures
- The producer reviews the final inspection report
- SQA group reviews the final inspection report

Follow-up: Validation/Verification - 2

- Data gathered during this activity
 - How much time was spent in follow-up
 - How many open issues were disputed



Follow-up: Exit Criteria

- Any change requests resulting from unresolved open issues have been submitted to the change approval process for handling
- The inspection report is completed and the producer agrees
- If necessary, a re-inspection is scheduled
- If necessary, issues are escalated to the project lead for resolution
- The inspection is noted as closed

Prevention Meeting



Prevention Meeting

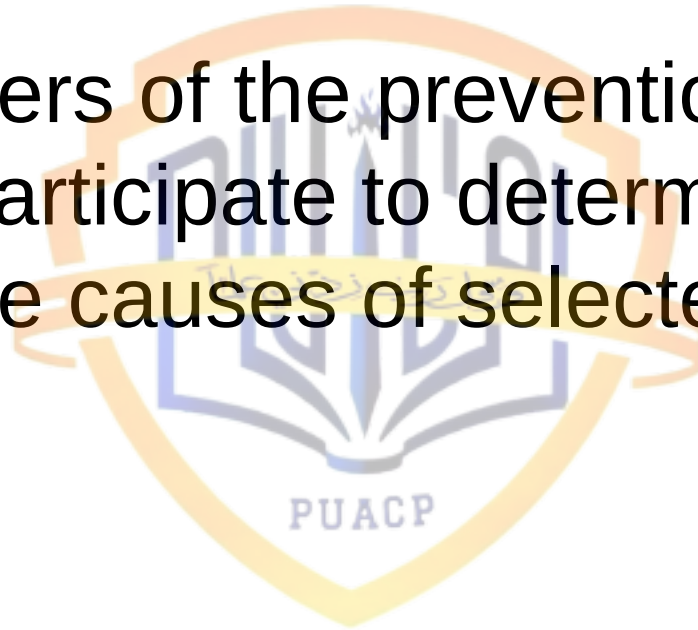
- Which is held periodically after sets of inspections have been performed to determine probable causes for selected defect types, instances, or patterns
- Required data about defects

Prevention Meeting: Responsibility

- The prevention team leader for the prevention meeting will record the results of the meeting and deliver proposals for actions to the organization management

Prevention Meeting: Other Roles

- The members of the prevention meeting team will participate to determine actions for probable causes of selected defect types



Prevention Meeting: Entry Criteria

- An inspection meeting was held
- The analysis meeting was held
- Defect data including causes are available to the prevention meeting team

Prevention Meeting: Tasks

- Record data from the prevention meeting
- Record proposed actions to be taken for defect prevention
- Initial preparation for the proposals to be presented to management for decision

Prevention Meeting: Validation/Verification - 1

- The prevention team has met based on the defined cycles for the meetings
- SQA reviews sampled reports
- The SEPG reviews proposed actions and resultant actions taken

Prevention Meeting: Validation/Verification - 2

- Data gathered during this activity
 - Time in the prevention meeting
 - Effort invested at the meeting
 - Number of proposals brought forward to management
 - Number of actions taken from proposals

Prevention Meeting: Exit Criteria

- The data is complete and agreed to by the prevention meeting participants





Data Recording and Reports



Data Recording and Reports

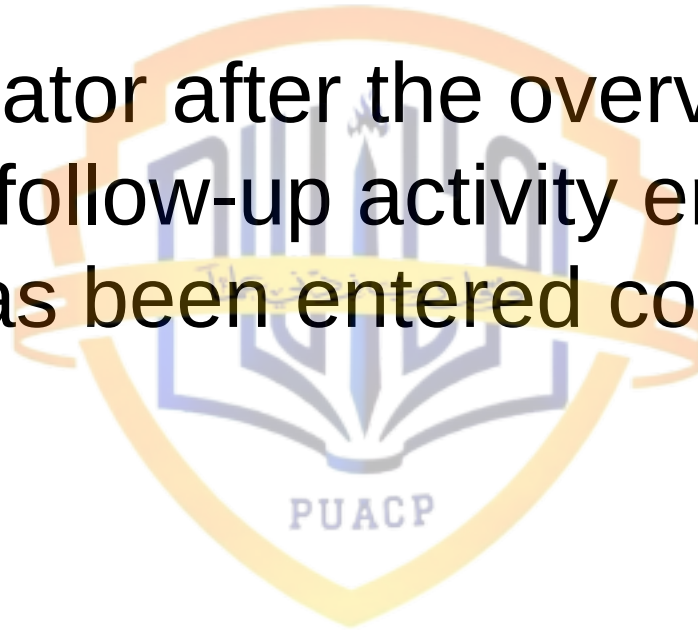
- To record the data about the defects and conduct of the inspection
- This activity is held concurrently with other activities, including at the end of the inspection process

Data Recording and Reports: Responsibility

- The recorder during the overview, inspection meeting, and optional analysis meeting records data about the defects and the conduct of the inspection
- Alternatively the moderator can enter the data

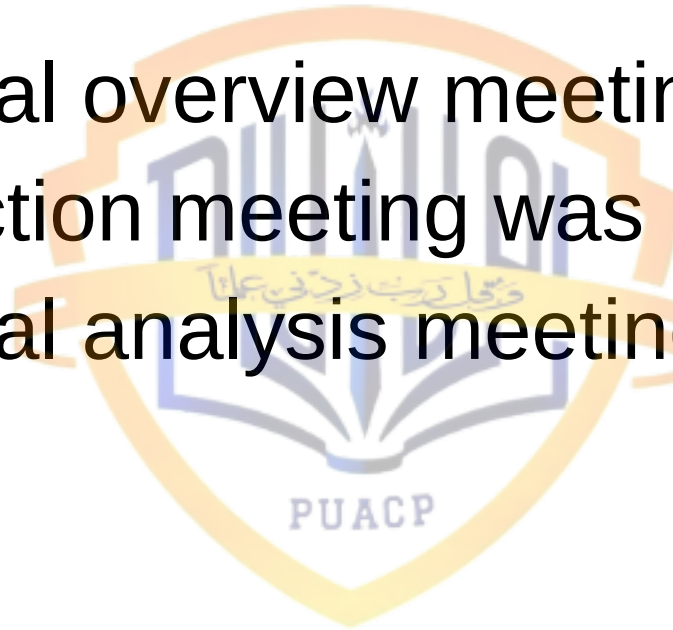
Data Recording and Reports: Other Roles

- The moderator after the overview and during the follow-up activity ensures that the data has been entered correctly and completely



Data Recording and Reports: Entry Criteria

- The optional overview meeting was held
- The inspection meeting was held
- The optional analysis meeting was held



Data Recording and Reports: Tasks

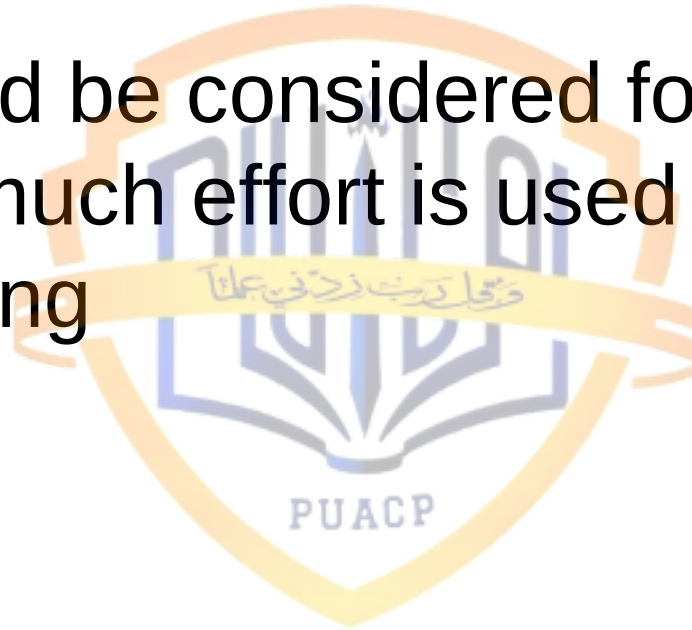
- Record data from overview, if held
- Record data at the inspection meeting, including preparation data
- Record data at the optional analysis meeting
- Record data during the follow-up activity, including sign-off to close the inspection

Data Recording and Reports: Validation/Verification - 1

- The inspection verifies the data at the end of the inspection meeting and optional analysis meeting
- SQA review sampled reports
- The producer reviews the report completed by the moderator

Data Recording and Reports: Validation/Verification - 2

- Data should be considered for this activity; e.g., how much effort is used for recording and reporting



Data Recording and Reports: Exit Criteria

- The data are complete and agreed to by the inspection meeting and analysis meeting participants
- The data gathered during the follow-up activity are complete and agreed to by the producer and moderator



Inspection Process Monitoring

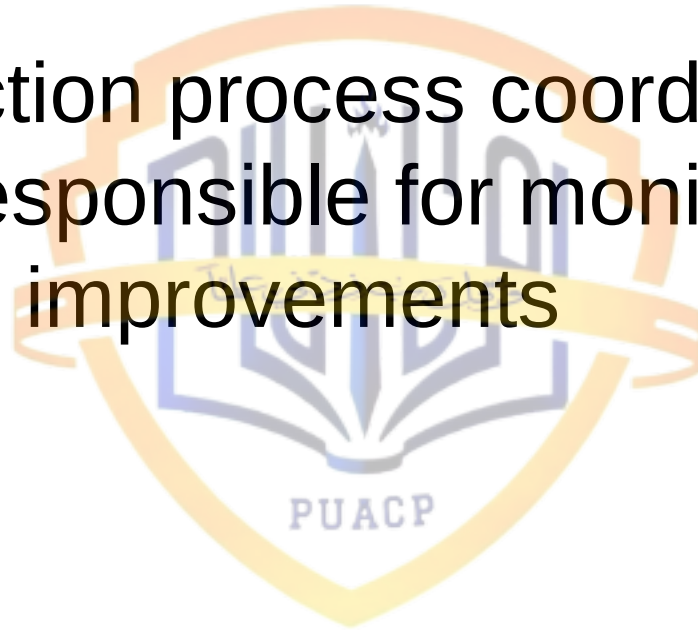


Inspection Process Monitoring

- This activity is held concurrently with other activities and after inspections
- The purpose is to evaluate the results of the inspection process as performed in the organization and to propose suggestions for improvement

Inspection Process Monitoring: Responsibility

- The inspection process coordinator or SEPG is responsible for monitoring and suggesting improvements



Inspection Process Monitoring: Other Roles

- Management ensures that inspection process monitoring is integrated into the inspection process
- The inspection process improvement team proposes actions for inspection process improvements based on the monitoring and analysis of the inspection coordinator

Inspection Process Monitoring: Entry Criteria

- Reports and results from inspections over a period of performance are available
- A coordinator is assigned
- Resources are allocated for inspection process improvement team

Inspection Process Monitoring: Tasks - 1

- Gather the inspection process data provided since the last monitoring report
- Review inspection reports and related data for trends and results against objectives
- Interview inspection participants to ensure understanding of results and to gather other inputs

Inspection Process Monitoring: Tasks - 2

- Perform analysis using data from the inspection reports, interviews, and surveys
- Provide the analysis to the inspection process improvement team for review and proposal to management for inspection process management

Inspection Process Monitoring: Validation/Verification - 1

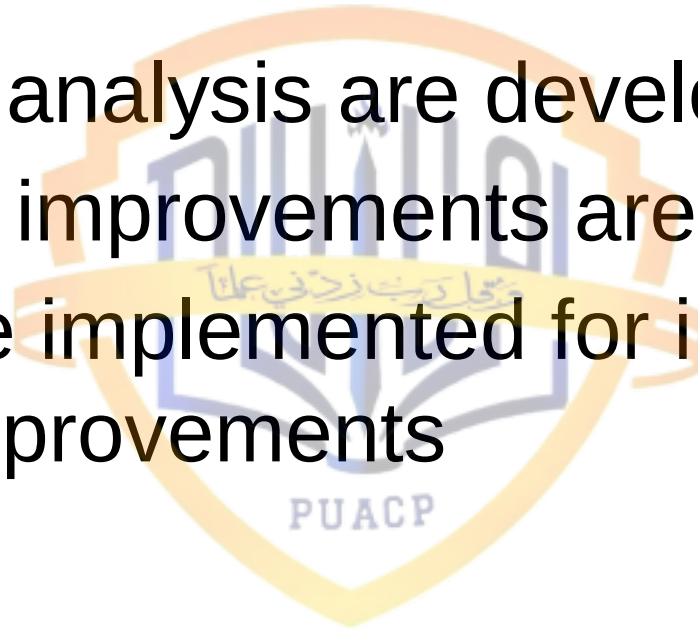
- The inspection coordinator performs monitoring actions per agreed periods for analysis
- The inspection process action improvement team meets per agreed periods for recommendations
- SQA reviews monitoring activity on a random basis to ensure it is being performed

Inspection Process Monitoring: Validation/Verification - 2

- Data gathered during this activity
 - How much effort is expended
 - How many proposals for improvement are made
 - How many improvements are put into action

Inspection Process Monitoring: Exit Criteria

- Reports of analysis are developed
- Actions for improvements are proposed
- Actions are implemented for inspection process improvements



References

- Software Inspections by Ronald A. Radice, Tata-McGraw Hill, 2003 (Chapter 3.8-3.12)

